
Olivia Chen

Computer Engineering Undergraduate

#618 27 N 6th St

San Jose, CA 95112

(510) 366-2750 | chenhuiya2005@gmail.com

LinkedIn: <https://www.linkedin.com/in/olivia-chen-118837280/>

OBJECTIVE

Computer Engineering undergraduate with hands-on experience in digital system design, Verilog RTL development, and low-level systems programming. Seeking a software or embedded systems internship to apply knowledge of operating systems, datapath design, FSM modeling, and hardware architecture to build efficient and reliable systems.

SKILLS

Programming Languages: Java, C, C++, Python, JavaScript, Verilog

Web Development: HTML, CSS, React, Git, Node

Tools: Linux (basic), Visual Studio Code, Vivado, GitHub, Google Workspace, Microsoft Office

Core Competencies: Object-Oriented Programming, Data Structures, Debugging, Technical Writing, Version Control, Collaboration, PCB Design/ HDL, FPGA Programming

PROJECTS & EXPERIENCE

GitHub - <https://github.com/Olivetree312>

- Includes repositories for exploratory coding projects in C++, Verilog, Javascript, asm, and recent coursework from CMPE 126: Algorithms and Data Structure Design
- Featuring: Array List & linked list implementations, recursion techniques, stack & queue structures, search algorithms

Personal Page (React-based) - <https://my-resume-site-azure.vercel.app/>

- Multi-page personal website built with React and deployed via Vercel
- Styled using HTML/ CSS, routed pages with React Router, hosted downloadable resume
- Version controlled using Git and GitHub with detailed commits

Projects

Assembly Code Simulator

- Built a simulator supporting arithmetic and logical instructions
- Simulated execution and control flow, demonstrated understanding of system-level programming

Data Structures Implementation (C++)

- Implemented linked lists, stacks, queues, and algorithms for data manipulation
- Applied pointer-based logic and memory management

Vending Machine FSM with Testbench, modeled on FPGA Board

- Designed and implemented vending machine Mealy FSM, programmed to Basys3 FPGA board
- Developed testbench to validate functionality and identify edge-case bugs
- Analyzed waveform using GTKWave and debugged timing issues
- Verified sequential logic and ensured correct state transitions

EDUCATION

San Jose State University, CA - *B.S. in Computer Engineering*

AUG 2023 - MAY 2027

Recent Courses:

- CMPE 50 - Object-Oriented Concepts and Methodology
- CMPE 126 - Algorithms and Data Structure Design
- CMPE 110 - Electronics for Embedded Systems
- CMPE 130 - Advanced Algorithm Design
- CMPE 125 - Digital Design II
- CMPE 102 - Assembly Language Programming
- CMPE 127 - Microprocessor Design I
- CMPE 142 - Operating Systems Design